

E-Commerce Data Analysis Report

Prepared by: Data Analytics Team

Company: ABC E-Commerce Pvt Ltd

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This report presents an in-depth analysis of sales performance, profitability trends, regional behavior, and discount impact using a structured data-driven approach. The objective is to uncover meaningful insights that help the business improve revenue, optimize pricing strategies, and strengthen customer engagement.

The analysis includes descriptive statistics, visual exploration, and strategic recommendations based on data patterns observed across different product categories, regions, and monthly cycles.

1. Descriptive Statistics

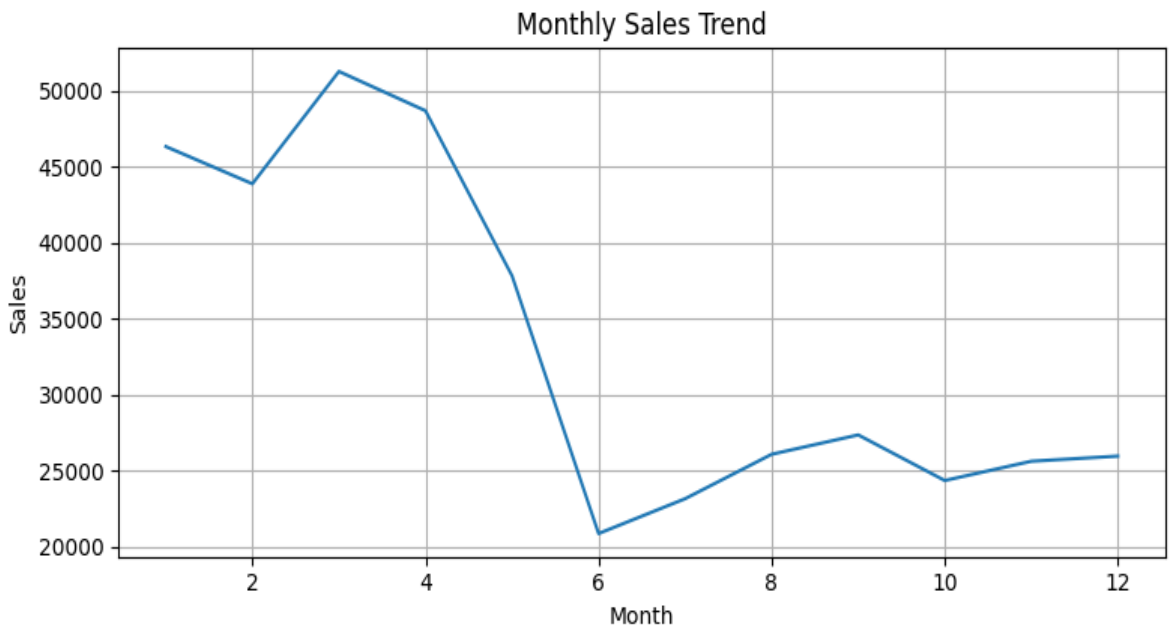
Average Sales: 803.02

Median Sales: 820.04

Average Profit: 152.34

Profit Variance: 40277.93

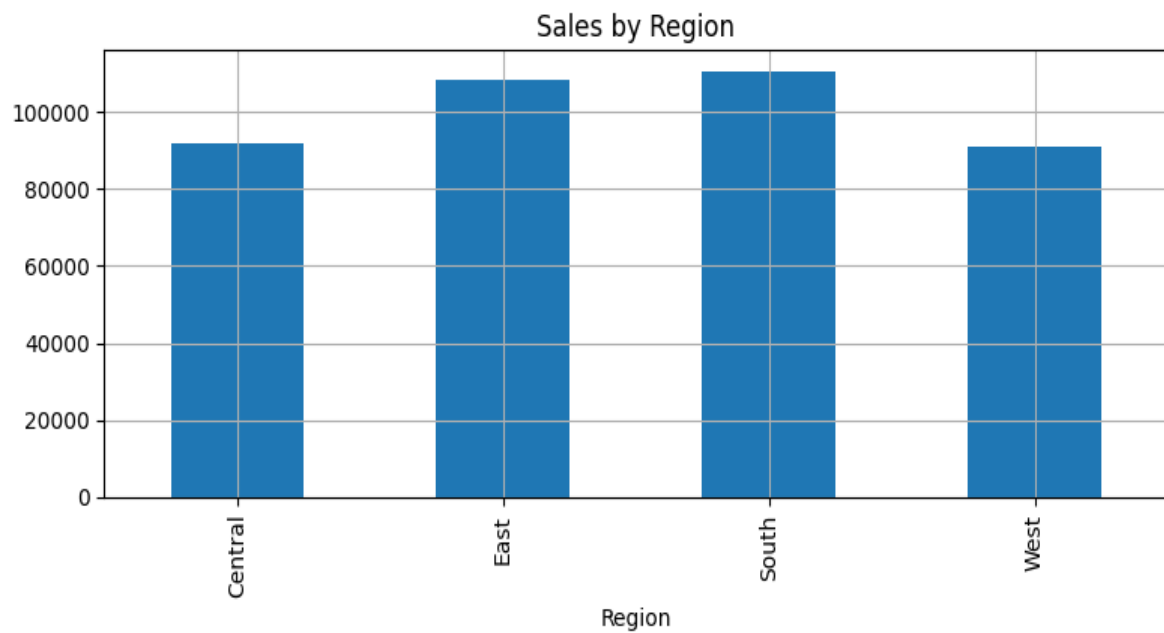
Monthly Sales Trend



The monthly trend highlights demand cycles and seasonal patterns. These insights help with budgeting, forecasting, and inventory decisions.

2. Regional & Category Performance

Sales by Region



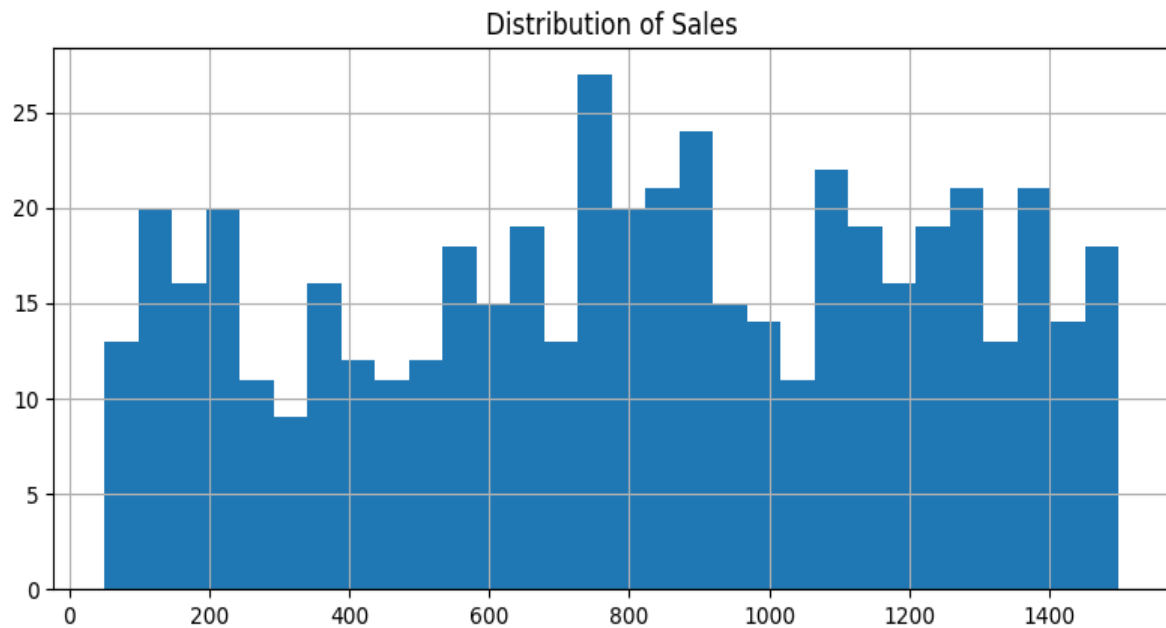
Regions showing higher sales represent stronger demand concentration. This enables targeted regional marketing initiatives.

Profit by Category

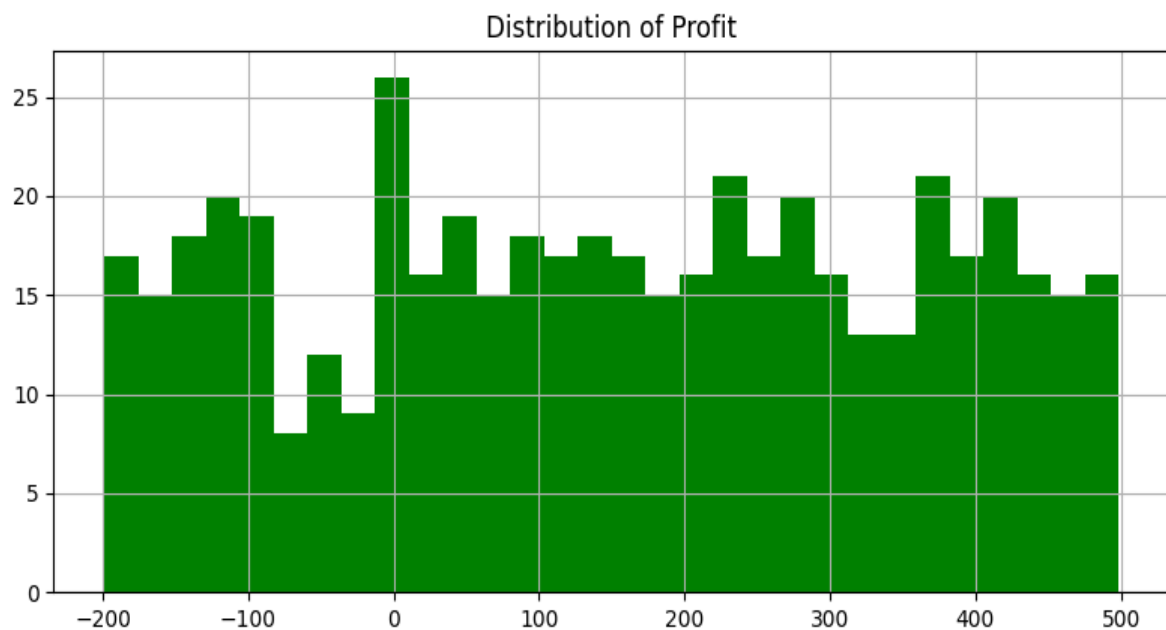


Category-level profitability guides decisions on pricing, product procurement, and catalog expansion.

3. Sales & Profit Distributions



Sales distribution exhibits right skewness, driven by a few high-value orders. This is typical in e-commerce.



Profit distribution identifies margin pressure points, especially where negative profit values appear.

4. Business Insights & Recommendations

- The best-performing region should receive prioritized investment, as it drives most sales.
- Low-profit categories require improved pricing, better cost negotiation, or discount reduction.
- Discounts significantly reduce profitability; revise discount rules and ensure minimum margin protection.
- Sales are highly skewed, suggesting opportunities for premium bundles and higher-value offerings.
- Marketing budgets and stocking strategies must align with the monthly demand cycles revealed in the data.